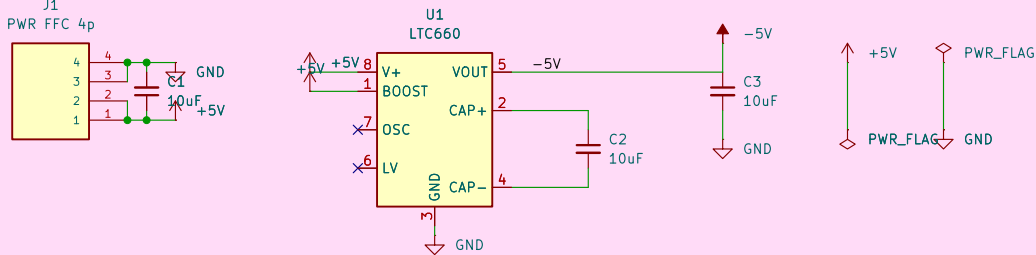


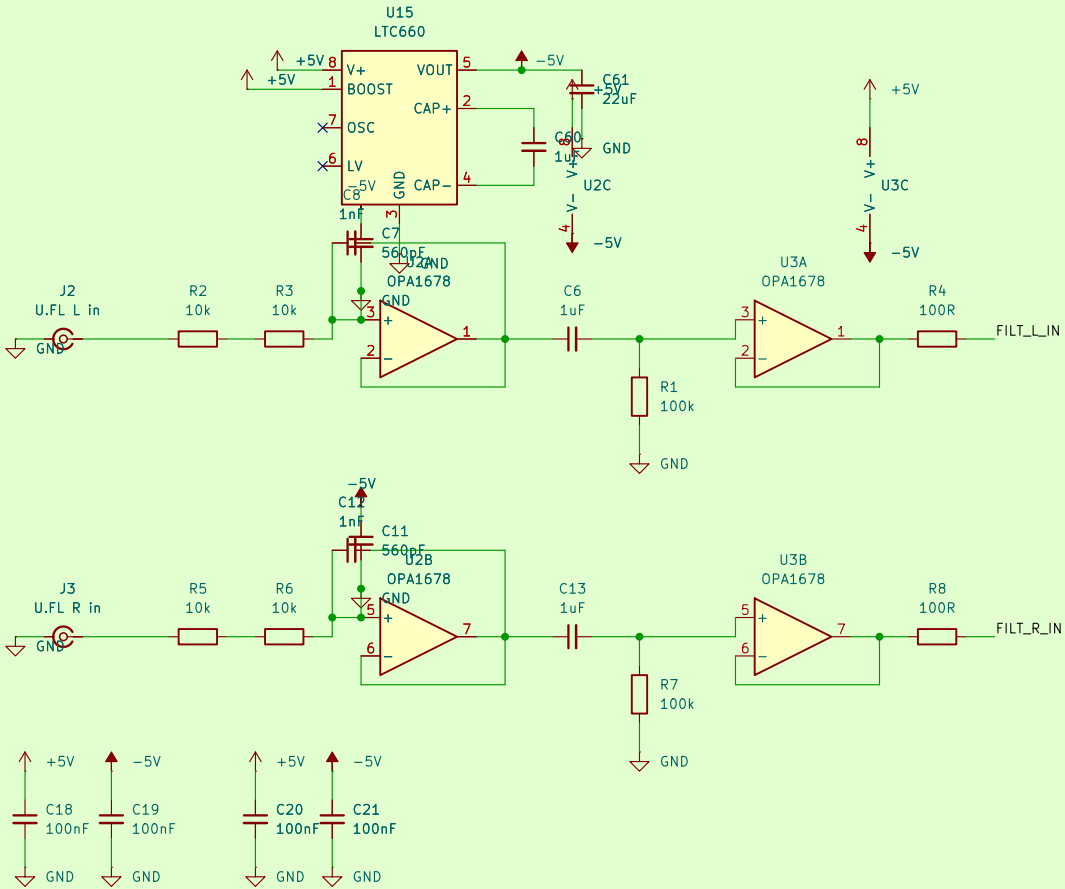
AUDIO BOARD – front end: power, U.FL audio in, reconstruction filter + buffers, 1/4in L/R + 3.5mm out

POWER INPUT
– 4-pin FFC from APM (+5V)
– ICL7660 makes -5V

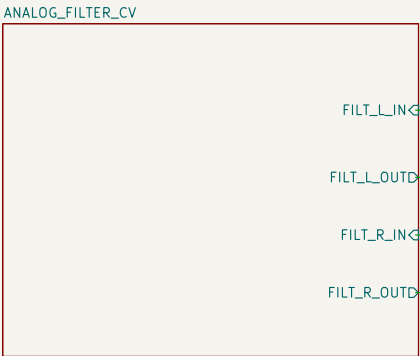


OUTPUTS – 1/4in L/R (ACJS-MHOM) + 3.5mm headphone/debug

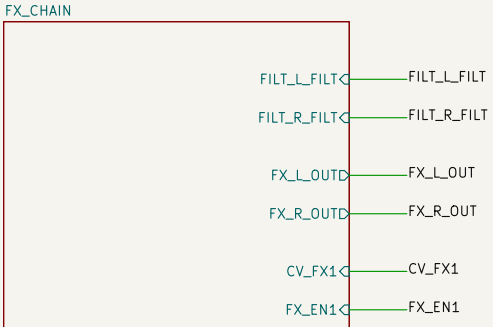
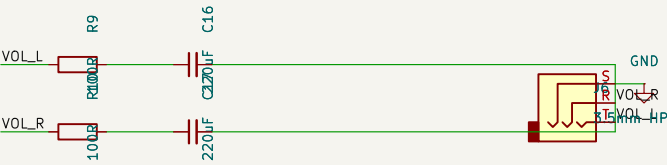
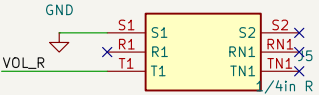
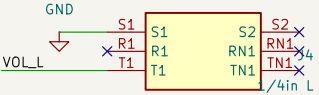
AUDIO INPUT
– U.FL for L and R DACs from APM



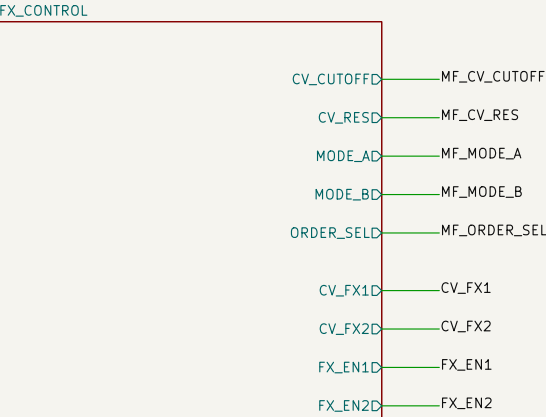
ANALOG PATH – each channel's buffer output (FILT_x_IN) goes into the analog sub-sheet; FILT_x_OUT returns to the jacks



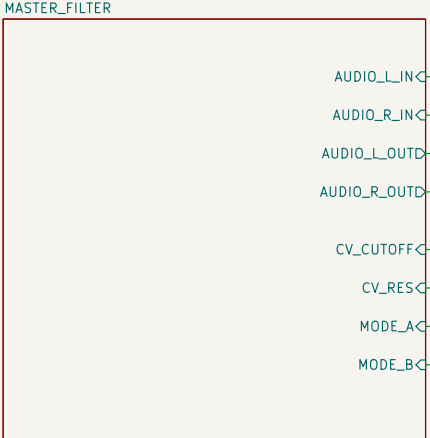
File: ANALOG_FILTER_CV.kicad_sch



File: FX_CHAIN.kicad_sch

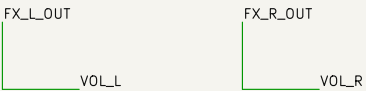


File: FX_CONTROL.kicad_sch



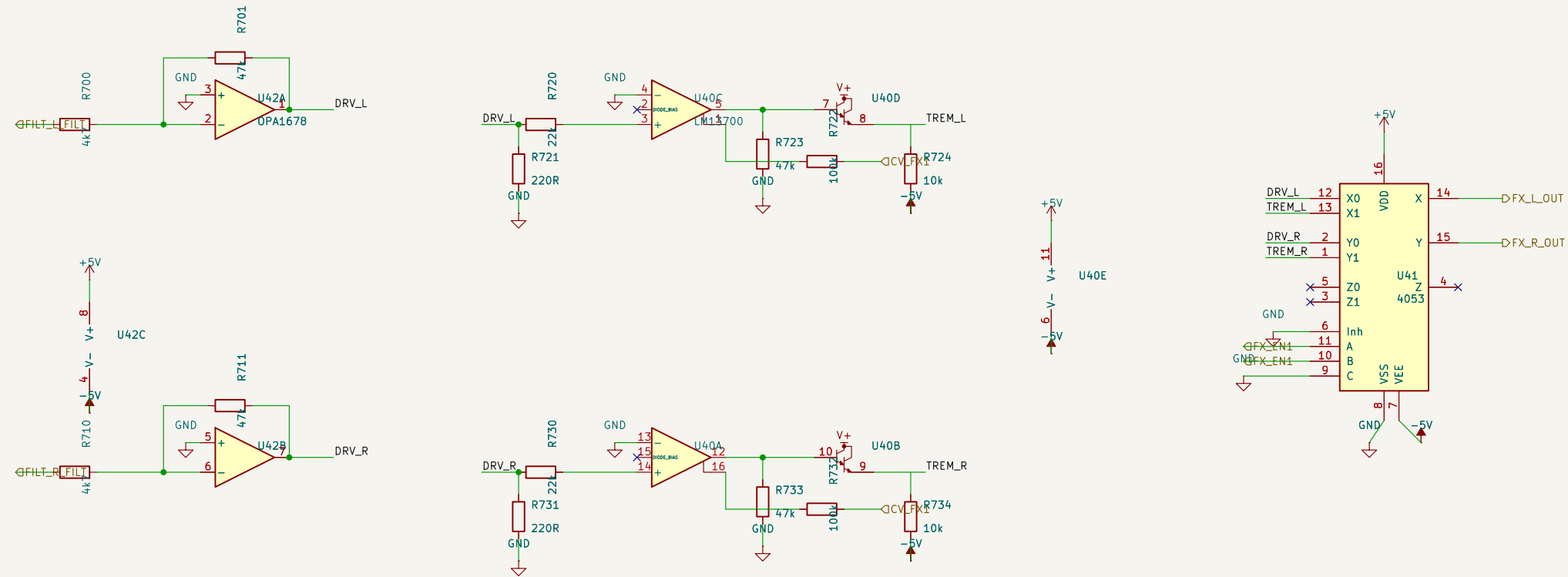
File: MASTER_FILTER.kicad_sch

MASTER VOLUME – RV4/RV5 = one dual-gang pot (shared L/R knob); attenuates final output to the jacks/HP



Sheet: /		
File: AUDIO_BOARD.kicad_sch		
Title:		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.4		
Id: 1/5		

FX CHAIN - ¹DRIVE (soft clip) + ²TREMOLO (OTA VCA, LFO=CV_FX1); each bypassable (FX_EN). In: ³FILT_x_FILT (from SVF) -> ⁴Out: ⁵FX_x_OUT (to volume)



DECODING 100nF/IC

C700 +5V -5V +5V -5V +5V -5V +5V -5V

GND 100nF GND 100nF GND 100nF GND 100nF GND 100nF GND 100nF GND 100nF

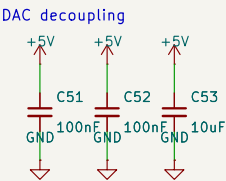
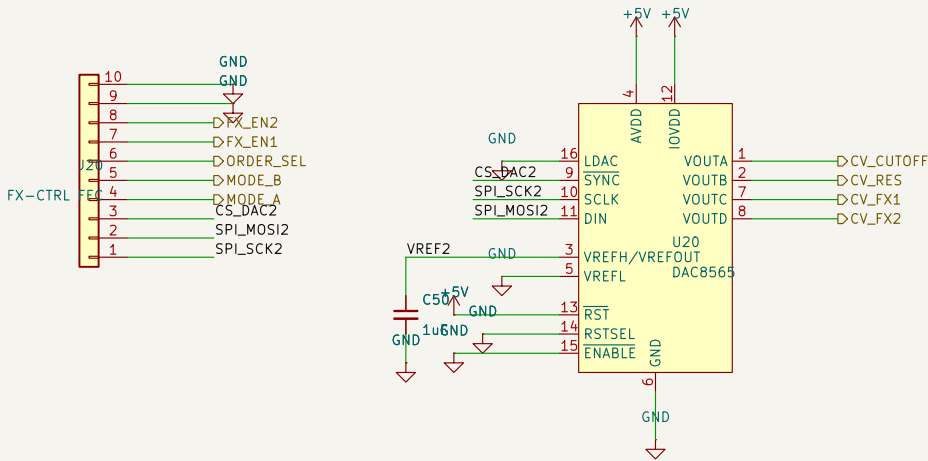
log FX ~ simulate/verify diode clip levels and OTA VCA vs LM13700 datasheet

NOTE: analog FX – simulate/verify diode clip levels and OTA VCA vs LM13700 datasheet before fab.

Sheet: /FX_CHAIN/		
File: FX_CHAIN.kicad_sch		
Title:		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 2/5

FX CONTROL – 2nd FX-control FFC (SPI + mode/order/enable) -> DAC8565 #2 makes CV_CUTOFF/CV_RES (+2 spare FX CVs); digital selects pass through

FX-CTRL FFC: 1SCK 2MOSI 3CS_DAC2 4MODE_A 5MODE_B 6ORDER_SEL 7FX_EN2 8FX_EN1 9VREF2 10GND

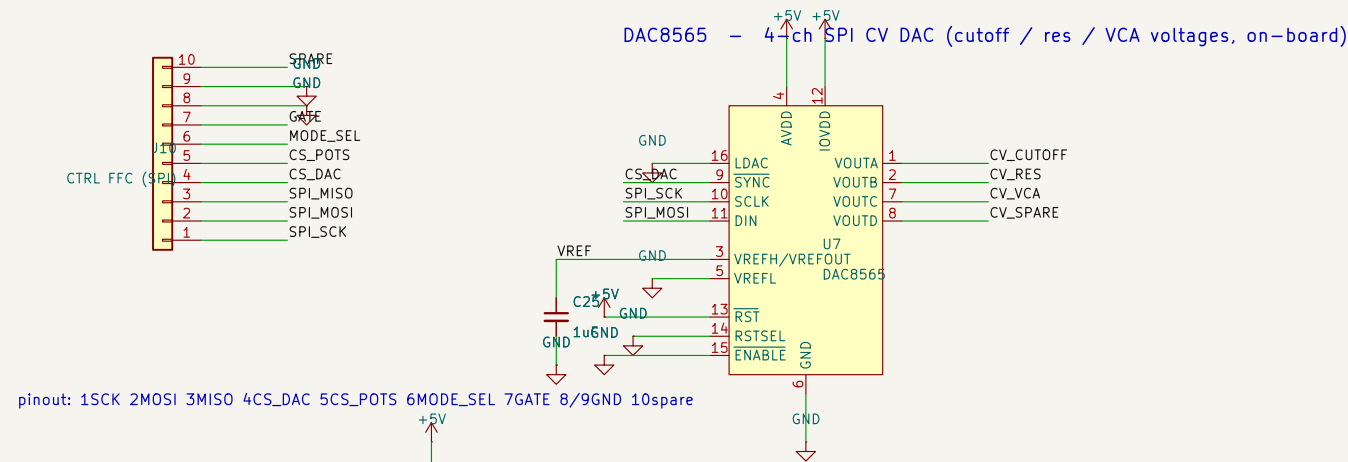


outputs to filter/FX: CV_CUTOFF, CV_RES (-> SVF), CV_FX1/CV_FX2 (-> future phaser/chorus/tremolo); MODE_A/B, ORDER_SEL, FX_EN1/2 digital.

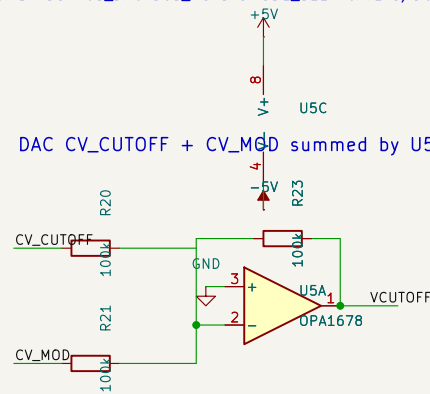
Knobs/switches live on the controller board; STM32 reads them, drives this DAC over SPI and sets the digital lines.

ANALOG FILTER / CV (stereo) – SPI in (10-pin FFC) -> DAC8565 makes CVs -> per-channel LM13700 VCF/VCA; CD4053 selects analog or FPGA-digital

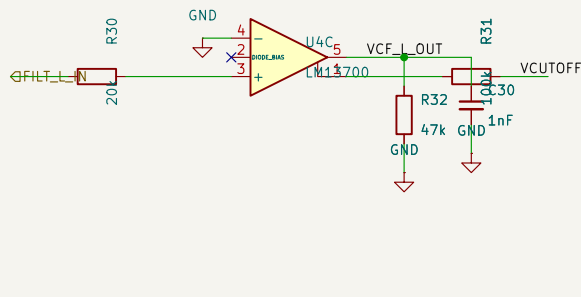
CONTROL – 10-pin FFC (SPI from APM)



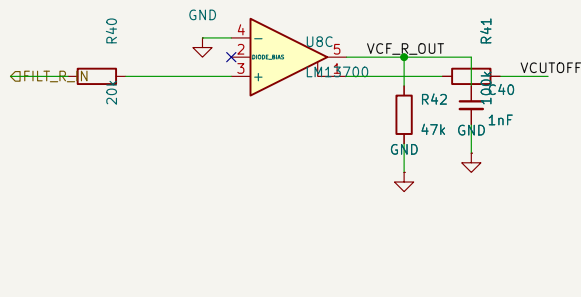
CV SUMMER – DAC CV_CUTOFF + CV_MOD summed by U5 -> VCUTOFF (drives both L and R VCFs); knob lives on the control board



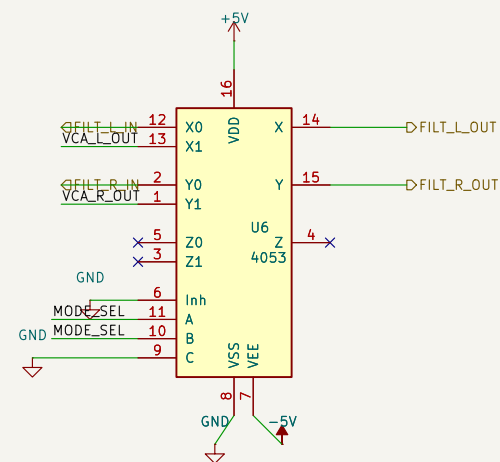
VCF+VCA L – LM13700 OTA lowpass then OTA VCA (NOTE: OTA/labc topology – verify vs datasheet)



VCF+VCA R – LM13700 OTA lowpass then OTA VCA (NOTE: OTA/labc topology – verify vs datasheet)



TOGGLE – CD4053: MODE_SEL selects analog (VCA_x_OUT) vs bypass (FILT_x_IN) for BOTH L and R

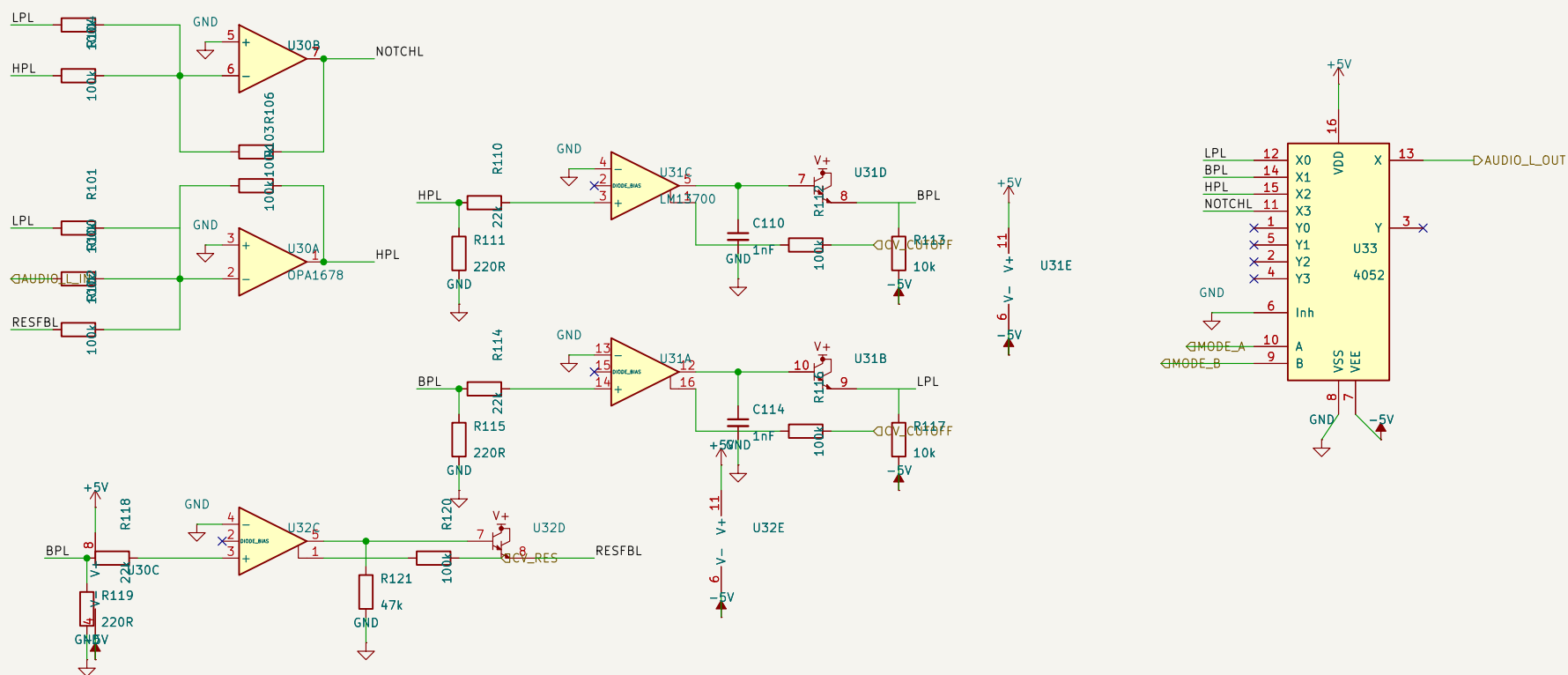


FILT_x_IN / FILT_x_OUT are the hierarchical pins shared with the root (front-end) sheet. Power rails (+5V/-5V/GND) are global.

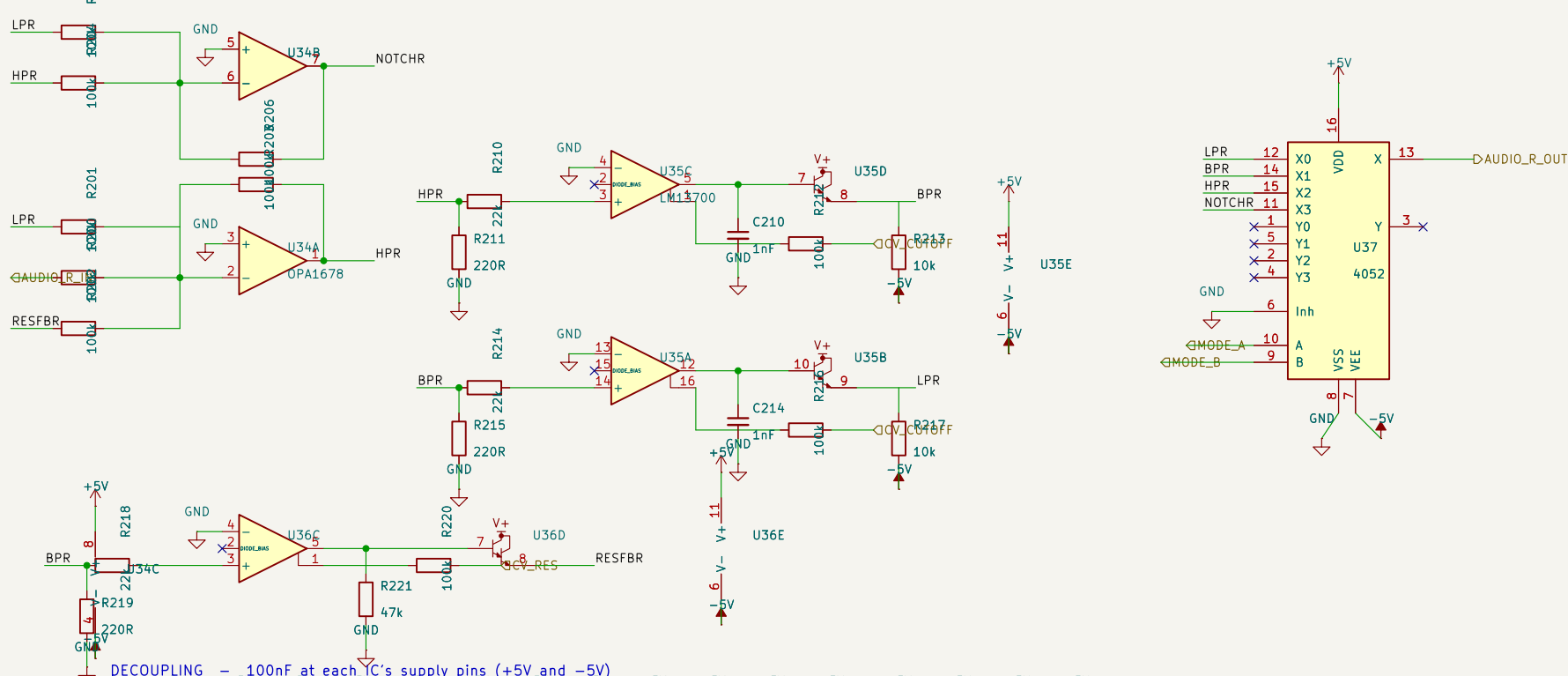
Sheet: /ANALOG_FILTER_CV/ File: ANALOG_FILTER_CV.kicad_sch		
Title:		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 4/5

MASTER FILTER (stereo) - VC state-variable filter 2-pole: LP/HP/BP/notch; CV cutoff + CV resonance; mode-select mux

CHANNEL L



CHANNEL R



DECOUPLING – 100nF at each IC's supply pins (+5V and -5V)

NOTE: OTA VC-SVF¹ + OTA resonance VCA² - prototype/verify levels; labc, buffer bias vs LM13700 datasheet before fab. ORDER_SEL reserved for the 2/4-pole cascade.⁴

Sheet: /MASTER_FILTER/ File: MASTER_FILTER.kicad_sch	
Title:	
Size: A3	Date:
KiCad E.D.A. 10.0.4	Rev: Id: 5/5